



Whole-genome evaluation and prophages characterization associated with genome of carbapenem-resistant *Acinetobacter baumannii* UOL-KIMZ-24-2

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Received: 21 September 2024 / Revised: 18 December 2024 / Accepted: 25 December 2024
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Abstract

Carbapenem-resistant *Acinetobacter baumannii* (CRAB) is an emerging threat to healthcare settings in many countries, principally in South Asia. The current study was aimed to identify, evaluate whole-genome and characterize the prophages in genome of CRAB strain, recovered from patients of Lahore General Hospital, Lahore. More than 200 samples were collected and identified by morphological and biochemical tests. These strains were also subjected to a comprehensive antimicrobial susceptibility evaluation using Kirby-Bauer method and further confirmed as CRAB strains by exploring *bla*_{OXA-51}. In addition, the whole-genome evaluation of a *Acinetobacter baumannii* UOL-KIMZ-24-2 was carried out using various Bioinformatics tools. A total of 150 strains of *A. baumannii* were recovered and identified in the current study. Among them, 49% strains were found resistant to carbapenem. The *bla*_{OXA-51} was found prevalent in the genome of *A. baumannii* recovered from medical ICU (38%). In addition, the UOL-KIMZ-24-2 genome analysis based on multilocus sequence typing (MLST) highlighted that UOL-KIMZ-24-2 belonged to ST2 (Pasteur scheme) sequence type. A total of 29 antimicrobial resistance (AMR) genes were present, importantly, *bla*_{OXA-66}, *bla*_{OXA-23} and *bla*_{OXA-25}. The mobile genetic elements (MGEs) were identified as transposases and belonged to four classes e.g. *IS15d1*, *ISAb24*, *ISEc29*, and *ISEc35*. A total of 14 virulence factors encoded by 58 different genes were detected in UOL-KIMZ-24-2. In addition, the phage sequences were identified in genome of UOL-KIMZ-24-2, divided into 3 regions. In conclusion, UOL-KIMZ-24-2 contained a mixture of AMR genes, MGEs, prophages sequences and virulence genes.

Keywords Antimicrobial resistance gene · Carbapenem-resistant *Acinetobacter baumannii* · Hospital-acquired infections · Prophage sequences

Communicated by Lydia Bogomolnaya.

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Introduction

Acinetobacter baumannii, a Gram negative, aerobic, non-fermenter coccobacillus, is a daunting emerging nosocomial (Blin et al. 2019) pathogen across the globe (Kurihara et al. 2020). The nosocomial infectious agent causes a wide spectrum of chronic infections, mostly evident in intensive care units (ICU) of hospitals (Chung et al. 2011; Ibrahim et al. 2021). The substantial risk factors of pathogenicity encompass immunosuppression, extended hospitalization, intrusive treatments, old age, and persistent broad-spectrum antimicrobial exposure (Chopra et al. 2013; Freire et al. 2016; Rai and Kumar 2022; Theriault et al. 2021). *Acinetobacter* species are found naturally resistant to many core antimicrobials, disinfectants, and desiccants (Diep et al. 2023), which advances long-term persistence and transmission in healthcare settings. However, extensive usage of broad-spectrum antibiotics favors antibiotic-driven resistance in pathogens, ultimately also leading to emergence of multi-drug resistant *A. baumannii*, accompanied by limitation of treatments for infected patients (Wareth et al. 2021).

The *Acinetobacter* infections are frequently treated by carbapenems but resistant against carbapenems in *A. baumannii* is spreading at previously unforeseen rates. For example, carbapenem resistance in *A. baumannii* was reported in range between 70 and 85% in the Asia-Pacific region (O'Donnell et al. 2021). Therefore, World Health Organization (WHO) has recognized *A. baumannii* as pathogen of concern, crucial for discovery and development of novel antimicrobial compounds, hence highlighting as a global threat to human health (Lee et al. 2014). Although, *A. baumannii* comparatively accounts for lower proportion of infections associated with Gram-negative pathogens, even though mortality rates associated with this infectious agent approach up to 40–50% in ICU (Freire et al. 2016). Evidently, *A. baumannii* have exhibited resistance to most of the commonly prescribed antibiotics, but polymyxin B is still considered valued as ultimate option for treatment of infections stemming from *A. baumannii* with carbapenem-resistance (Nang et al. 2021). However, resistance genes against polymyxin B have also been identified in carbapenem-resistant *A. baumannii* (CRAB), which is already characterized as difficult to treat due to availability of limited relevant drugs in medical practice (Martins-Sorenson et al. 2020).

CRAB poses a significant threat to human health along with persuasive concerns in healthcare settings. CRAB isolates, being extensively resistant to broad spectrum antimicrobial agents, majorly targets hospitalized patients, and triggers multi-drug resistance (MDR) associated infections contributing to high mortality rate (Asif et al. 2018). Fundamentally, *A. baumannii* exhibits carbapenem resistance

by overexpressing chromosomal-mediated and/or plasmid-oxacillinases encoding by *bla*_{OXA} genes and existence of metallo-carbapenemases (*bla*_{IMP}, *bla*_{VIM}, *bla*_{SIM}, and *bla*_{NDM}) (Pagano 2016; Poirel and Nordmann 2006). Importantly, two major clones of *A. baumannii*, GC1 and GC2 which were reported resistant to older during 1970s, they have now demonstrated resistance even to advanced drugs such as cephalosporins, fluoroquinolones, and carbapenems (Holt et al. 2016). The majority of CRAB strains, carrying *bla*_{OXA-23} associated with GC2 type, have been reported worldwide. GC2 clone of *A. baumannii* has been predominantly disseminated in Europe (Van Dessel et al. 2004), Australia (Higgins et al. 2010), Asia (Kim et al. 2013) and United States (Adams et al. 2019).

Phages infect bacteria naturally and considered as natural bacterial killers (Wang et al. 2024). Due to an increased emergence of antibiotics resistance in pathogenic bacteria, phages are considered as valuable and promising alternatives to antibiotics for eliminating pathogenic bacteria worldwide (Zhang et al. 2022). Recent studies showed great therapeutic potential of phages against pathogenic *A. baumannii*. For example, a successful phage-based therapy against multidrug-resistant (MDR) *A. baumannii* infection in diabetic patients was reported by Schooley et al. (2017). Previous studies by various researchers have now confirmed that MDR *A. baumannii* infections can be treated effectively by phages with no deleterious side effects (Torabi et al. 2021; Sisakhtpour et al. 2022). However, after infecting *A. baumannii* strains, phages could become part of the genome of *A. baumannii*, hence start multiplying along bacterial own cell division. These prophages might then be involved in transferring AMR genes to other bacteria by generalized transduction (Stanczak-Mrozek et al. 2017).

The current study was aimed to assess the prevalence of *A. baumannii* strains in patients of Lahore General Hospital, Lahore and to characterize morphology and biochemical aspects of the isolated *A. baumannii* strains. The research work also evaluated the antimicrobial susceptibility profile of the isolated *A. baumannii* strains and confirmed the carbapenems resistance by detecting *bla*_{OXA-51} gene. In addition, a prominent CRAB strain, UOL-KIMZ-24-2, was subjected to whole-genome evaluation in the present study. In the in silico analysis, the genome of *A. baumannii* UOL-KIMZ-24-2 was studied to reveal various genetic factors e.g. AMR, MGE genes etc. contributing toward antibiotics resistance in UOL-KIMZ-24-2. The research work also reported presence of prophage regions in the genome UOL-KIMZ-24-2 consisting of 3 different regions.

Materials and methods

Study setting

Lahore General Hospital (LGH) is one the major teaching hospital in Lahore over an area of 256 Kanals located on Ferozepur Road, Lahore, Pakistan. The hospital, equipped with over 1600 beds, facilitates approximately 1,250,540 outdoor patients/year, 78,000 indoor patients/year, 1,103,423 emergency patients/year and 45,000 surgical procedures/year. The LGH features 22 basic and clinical departments, with an Intense Care Unit (ICU) accommodating over 100 beds. Most of the patients involved in this study were from LGH and nearby localities.

Study design

A total of 200 samples were collected from various sources and locations within the medical facility over a period of time (2022–2023). The samples include endotracheal tube tips, urinary catheter tips, tracheal secretions, wound swabs, blood, pus swabs, sputum, drain tips, cerebrospinal fluid, and urine. These samples were collected from different wards and units at LGH such as medical intensive care unit (MICU)-LGH, Paeds Ward-LGH, Surgery Ward-LGH, Paeds ICU-LGH, Surgery ICU-LGH, Out Patient Department (OPD)-LGH, Neonatal ICU-LGH, Burn Unit-LGH, Gynae Ward-LGH, Medical Ward-LGH, and Neurology Ward-LGH (Table S1). A written consent from patients' guardians/caregivers were acquired prior to enrollment of patient in the respective study. The information related to microbiological data such as type of sample (pus/wound swab, urine, blood, sputum/BAL), date and place of positive culture and sample collection, results of antimicrobial susceptibility testing (AST) and clinical diagnosis was collected. The data was collected in a case report form from all study participants and subsequently transferred to an electronic database.

Recovery of *Acinetobacter* species from collected samples

The acquired samples were cultured in laboratory on MacConkey agar, Blood agar and Chocolate agar and were subjected to incubation at 37 °C for 24 h. Recovered clinical isolates were identified through conventional microbiological techniques. Temporarily, all recovered isolates were preserved in 0.8% nutrient agar deeps at stored at 4 °C. The isolates were inoculated in Tryptic Soy Broth (TSB) and stored at -80 °C for long term storage, while significant isolates were lyophilized to ensure stability.

Morphological and biochemical evaluation

A. baumannii species were typically characterized through colony morphology, biochemical tests and Gram staining (Alsan and Klompas 2010). The shape, arrangement, motility and Gram characteristics of bacterial strain were observed through microscopic examination at different magnifications. Furthermore, all clinical isolates were identified as *Acinetobacter* species through biochemical tests such as indole, citrate, oxidase, urease, catalase, methyl red, triple sugar iron and Voges-Proskauer (Bergey 1994; Holt et al. 2016). Moreover, phenotypic identification of the selected isolates was carried out through sugar fermentations tests (glucose, lactose, mannitol, adonitol, dulcitol, melibiose, esculin, decarboxylases as lysine, arginine and ornithine) according to standard guidelines described for API 20E.

Antimicrobial susceptibility testing

The antibiotic resistance patterns of *A. baumannii* isolates against various classes of antibiotics was determined by Kirby-Bauer method (Begum et al. 2013). The respective antimicrobials including amikacin, ampicillin-sulbactam, cefepime, ceftazidime, cefotaxime, ceftriaxone, ciprofloxacin, cotrimoxazole, doxycycline, gentamicin, imipenem, meropenem, levofloxacin and piperacillin-tazobactam. American Type Culture Collection (ATCC®) strains, *Pseudomonas aeruginosa* ATCC® 27,853 and *Escherichia coli* ATCC® 25,922, served as quality controls. Antimicrobial Susceptibility of tested isolates was evaluated according to the Clinical Laboratory Standard Institute (CLSI) guidelines (2019) (CLSI 2021). Frequency (percentage) of data was calculated to summarize the percentage of *A. baumannii* colonization, prevalence of clinical cultures infection, comorbidities, and ICU outcomes included in the respective study. MS Excel XP® (Microsoft®, Northampton, MA, USA) was used for storage and analysis of obtained data.

Detecting *bla*_{OXA-51} in *A. Baumanii*

OXA-type carbapenemases encoding *bla*_{OXA} lineage genes were identified to elucidate the presence of carbapenem resistance (Woodford et al. 2006). The multiplex polymerase chain reaction (PCR) with specifically designed primers for *bla*_{OXA-51} was used using specific forward and reverse primers as described by Brown et al. (2005). Amplified PCR products were confirmed in 1% agarose gel (Sigma-Aldrich, United States).

Whole genome sequencing of *A. Baumannii* UOL-KIMZ-24-2

DNA extraction, library construction and sequencing

Among the 150 strains, *A. baumannii* H-KIMZ-24-2 was selected further for whole-genome analysis based on its resistant to multiple antibiotic classes. Genomic DNA extraction of UOL-KIMZ-24-2 was carried out through DNeasy blood and tissue kit (Qiagen, Germantown, MD). The extracted DNA was sent to MicrobesNG United Kingdom for whole genome sequencing. Library was prepared with Nextera XT library preparation kit and complete genome of UOL-KIMZ-24-2 was sequenced on the Illumina Miseq using a 2 × 250 bp paired-end configuration.

Genome assembly and annotation

Initially, FastQC v0.11.5 was employed for quality assessment of short Illumina reads with default configuration. The low quality reads, contamination and adapter sequences were discarded by using Trimmomatic version 0.38 (Bolger et al. 2014). Flye v2.7 was used for generation of chromosomal assembly (Kolmogorov et al. 2019). Short-reads sequencing data was processed for plasmid recovery, and subjected to de novo assembly through SPAdes v3.13.0 (Bankevich et al. 2012). The obtained whole-genome sequence was refined and polished using Pilon v1.23 (Walker et al. 2014). Quality of refined assembly was evaluated through BUSCO v4.0.6 (Simão et al. 2015), referring pseudomonales_odb10 as database. Finally, the sequenced genome was submitted to NCBI/Gene Bank for getting accession number.

Bioinformatics analysis

Several Bioinformatics tools were used to analyze the obtained sequences of *A. baumannii* UOL-KIMZ-24-2. Briefly, general genomic statistics of UOL-KIMZ-24-2 was assessed using DFAST server (Tanizawa et al. 2018). The functional genomic analysis of UOL-KIMZ-24 was carried out using MG-RAST (Keegan et al. 2016). In addition, virulence factors / genes were annotated in the genome of UOL-KIMZ-24-2 using Virulence Factor Database (VFDB) (Liu et al. 2022). The ABRicate v1.0.1 using comprehensive antibiotic resistance database (CARD) was used to predict the antibiotic resistance genes (AMR) (Alcock et al. 2020). The mobile genetic elements (MGE) associated the AMR were predicted in the genome of UOL-KIMZ-24-2 using VRprofile2 (Wang et al. 2022). Prophage regions identified in the genome of UOL-KIMZ-24-2 through PHASTER database (Wishart et al. 2023).

Taxonomic classification and phylogenetic analysis

The sequenced genome of UOL-KIMZ-24-2 was deposited into NCBI Gene Bank for getting the accession number. Moreover, phylo-genomic analysis was carried out using Type Strain Genome Server (TYGS) (Meier-Kolthoff and Göker 2019). The closely related type strain was determined in two steps. Initially, the user strain was analyzed with all available type strains in TYGS by using MASH algorithm followed by selected ten type strains with least MASH value. Secondly, 16 S rRNA gene was extracted via RNAmmer (Lagesen et al. 2007), BLASTed (Camacho et al. 2009) and compared with ten closely related TSs (Meier-Kolthoff et al. 2013). The FASTME 2.1.6.1 including SPR post-processing were used to infer evolutionary tree with branch support of resulted intergenomic distances (Lefort et al. 2015). 100 pseudo-bootstrap replicates were used to support the branches in the tree. The trees were visualized with PhyD3 (Kreft et al. 2017) and entrenched at the midpoint (Farris 1972).

Results

A total of 150 strains of *A. baumannii* were recovered from 200 samples, collected from various sources and locations within the medical facility over a period of time (2022–2023) (Table S1).

Morphological and biochemical evaluation

A. baumannii species were typically identified through colony morphology, Gram staining, and biochemical tests. Colony characteristics were noted based on their shape, size, color, opacity, and mucosity. Morphologically, *A. baumannii* appeared as Gram-negative, coccobacilli with rounded ends, often arranged singly or in pairs. They exhibited a relatively small size, ranging from 1 to 2 micrometers in width and 1–3 micrometers in length. Growth occurred readily on standard culture media within a temperature range of 12 to 43 °C, with an optimal temperature around 37 °C, and at a pH range of 7.4 to 7.6 (Fig. 1).

Moreover, all clinical isolates were identified as *Acinetobacter* species by biochemical tests (Table 1). The isolates were catalase and citrate positive. Likewise, sugar fermentations tests (glucose, lactose and mannitol, adonitol, dulcitol, melibiose and esculin, decarboxylases (lysine, arginine and ornithine) were conducted for phenotypic identification of the selected isolates as per standard guidelines described for API 20E. The API 20E tests confirmed isolated as *Acinetobacter* species.

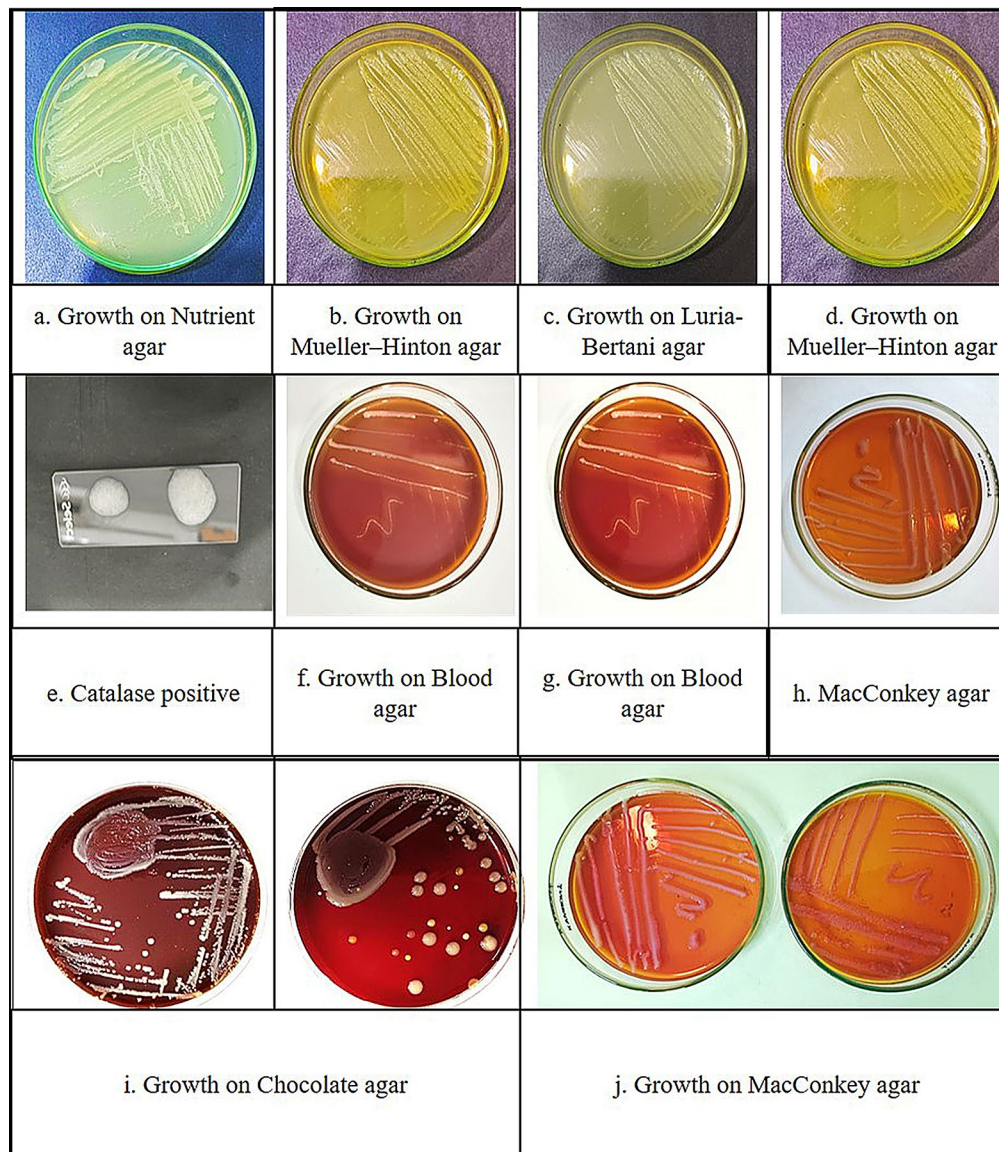


Fig. 1 Growth Characteristics of *A. baumannii* strains on various agar media

Table 1 Biochemical and phenotypic characteristics of *A. baumannii* species

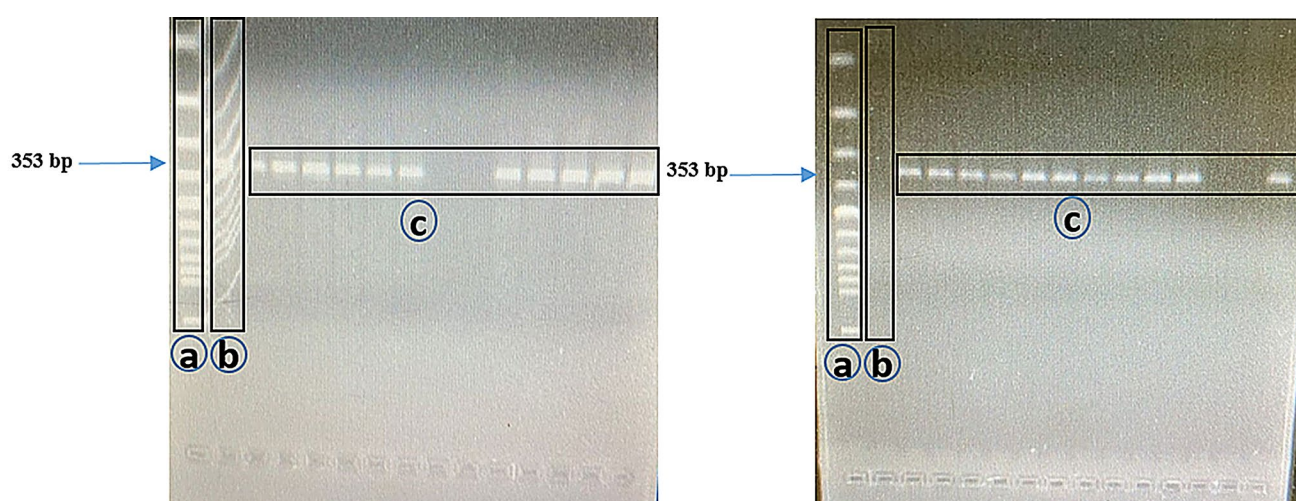
Tests	Characteristics
Gram stain	Gram negative coccobacilli
Oxidase	-
Catalase	+
Indole	-
Methyl Red	-
Voges proskauer	-
Citrate	+
Urease	-
Triple sugar iron	Alkaline slant / alkaline butt
Motility	Non motile

Antibiotic susceptibility testing

Assessment of antimicrobial susceptibility against 14 antibiotics was performed for all 150 *A. baumannii* strains. The antibiotic resistance pattern of isolated *A. baumannii* strains is displayed in (Table 2). Overall, a total of 118 strains showed MDR profile(s) as they were resistant to at least one agent in 3 or more antibiotics categories (Magiorakos et al. 2012). Notable findings include high resistance to certain antibiotics such as piperacillin-tazobactam (78.6%) and ciprofloxacin (74.0%). In addition, resistance percentages were found vary across the antibiotics classes e.g. resistance to carbapenems e.g. meropenem were found in 49.3% strains. Amikacin shows relatively higher resistance at 70.0%.

Table 2 Antibiotic resistance pattern of isolated *A. Baumannii* evaluated as per CLSI guidelines (2019)

S. No.	Family of antibiotics	Antibiotics	Abbreviation	No of resistant strains	%age
1	Penicillins	Ampicillin-sulbactam (10/10µg)	SAM	87	58.0
2		Piperacillin-tazobactam (100/10µg)	TZP	118	78.6
3	Cephalosporins	Ceftazidime (30 µg)	CAZ	112	74.6
4		Ceftriaxone (30 µg)	CRO	116	77.3
5		Cefotaxime (30 µg)	CTX	80	53.3
6		Cefepime (30 µg)	FEP	87	58.0
7	Carbapenems	Imipenem (10 µg)	IMP	18	12.0
8		Meropenem (10 µg)	MEM	74	49.3
9	Aminoglycosides	Gentamicin (10 µg)	CN	73	48.6
10		Amikacin (30 µg)	AK	105	70.0
11	Fluoroquinolones	Levofloxacin (5 µg)	LEV	76	50.6
12		Ciprofloxacin (5 µg)	CIP	111	74.0
13	Tetracyclines	Doxycycline (30 µg)	DO	29	19.3
14	Sulfonamides	Cotrimoxazole (25 µg)	SXT	87	58.0

**Fig. 2** The photographs show positive confirmation of *bla*_{OXA-51} like gene (353 bp). (a) 100 bp DNA ladder (b) negative control (c) Confirmation of *bla*_{OXA-51} in *A. baumannii* isolates

Gentamicin and cotrimoxazole exhibited moderate resistance at 48.6% and 58.0%, respectively (Table 2).

Detecting *bla*_{OXA-51} in *A. Baumannii*

In the study, *bla*_{OXA-51} was PCR amplified and genotyping of *A. baumannii* isolates was carried out to confirm presence of *bla*_{OXA-51} genes before sending *A. baumannii* species for whole genome sequencing. Presence of a 350 bp band indicates presence of *bla*_{OXA-51} in the genome of *A. baumannii* (Fig. 2).

In addition, the *bla*_{OXA-51} was found prevalent in the genome of *A. baumannii* recovered from medical ICU (38%) followed by surgical ICU and OPD (Fig. 3). While the main source of *bla*_{OXA-51} was found in *A. baumannii* isolated from endotracheal tube tip (20.6%) followed by tracheal secretions (14.6%) and wound swabs (14%) (Fig. 3).

Whole genome sequencing of *A. Baumannii* UOL-KIMZ-24-2

Genomic characteristics / properties

The draft genome of UOL-KIMZ-24-2 consists of 140 contigs, 3,935,051 bp genome size (with a GC content of 38.9%), N50 of 181,291, 3,865 coding sequences (CDSs) with an estimated coding ratio of 86.9%, 66 tRNAs, and 7 rRNA, whereas no CRISPR arrays were observed in the genome of strain UOL-KIMZ-24-2 (Table 3). Based on MLST analysis, UOL-KIMZ-24-2 belonged to ST2 (Pasteur scheme) sequence type (Table 3).

In addition, the functional genomic study revealed that this genome contained 418 subsystems (Fig. 4). Overall, 45% of subsystems were recognized to encode hypothetical proteins and rest of 55% were found to perform known functions (Table 4).

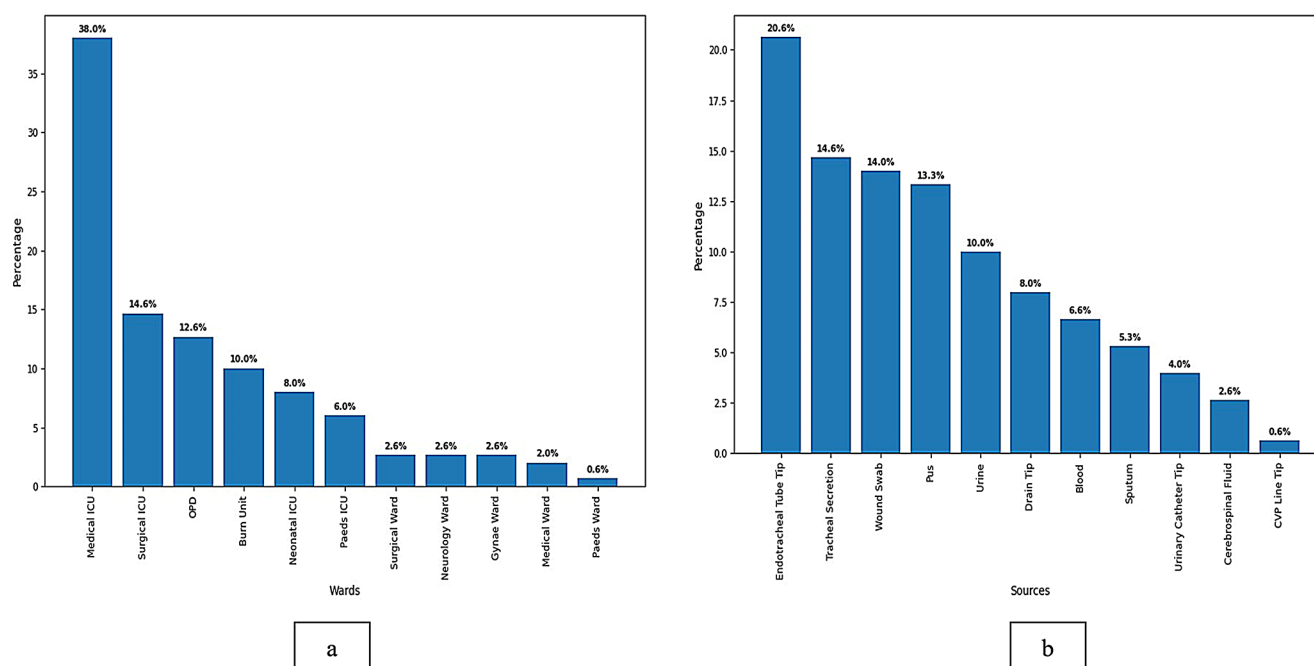


Fig. 3 Prevalence of *bla*_{OXA-51} in the genome of *A. baumannii* in various (a) wards (b) sources

Table 3 Functional genomic properties / features of UOL-KIMZ-24-2 using MG-RAST server

Genomic feature	Value
Size (bp)	393,505
GC Content (%)	38.9%
Number of Contigs	140
Number of Coding Sequences (CDS)	3849
Number of RNAs	83
Antibiotic Resistance Genes	29
Virulence Genes	58
Prophage regions	3
CRISPR arrays	0

This known fraction of genomic data was sub-categorized further into 22 subsystems namely as RNA metabolism (110), nitrogen metabolism (28), cell division and cell cycle (31), stress response (114), nucleosides and nucleotides (76), phosphorus metabolism (39), carbohydrates (266), prophages, transposable elements and plasmids (18), metabolism of aromatic compounds (107), secondary metabolites (5), DNA metabolism (81), potassium metabolism (18), respiration (85), dormancy and sporulation (2), miscellaneous (54), amino acids and derivatives (398), sulfur metabolism (58), iron acquisition and metabolism (27), membrane transport (106), respectively.

Taxonomy and phylogenomic analysis

The accession number achieved from NCBI/Gene Bank is JBJHCQ000000000 (First version). The phylogenetic analysis revealed that the strain UOL-KIMZ-24 clustered with

A. bamuannii and showed maximum similarity up to 100% based on 16 S rRNA gene and whole genome sequences using Type Strain Genome Server (TYGS) (Fig. 5A and B).

Furthermore, the ANI results indicated that UOL-KIMZ-24-2 showed 99.47% similarity with *Acinetobacter baumannii*, showing close relatedness with the reference genome (Table 5). Thus, these findings inferred that strain UOL-KIMZ-24-2 is an *A. bamuannii* strains.

Antimicrobial resistance (AMR) determinants

The prediction of antimicrobial resistant genes was evaluated by CARD database in UOL-KIMZ-24-2 genome highlighted that strain harbored number of determinants responsible for resistance to broad spectra of antimicrobial classes (Fig. 6). A total of 29 antimicrobial resistance (AMR) genes were identified on the chromosome of *A. baumannii* strain UOL-KIMZ-24-2 (Table S2). The genes like *bla*_{OXA-66}, *bla*_{OXA-23} and *bla*_{OXA-25} were detected in the genome which belonged to beta-lactum class and has high resistance for imipenem, meropenem and other antibiotics belonging to beta-lactum class.

Besides them, a few genes are involved in imparting resistance against amikacin, gentamicin, tobramycin, isepamicin, netilmicin and streptomycin class antibiotics by *aph(6)-Id*, *aph(3)-Ib* and *armA*. Additionally, two genes namely *msr(E)* and *mph(E)* that represent resistance to antibiotics including erythromycin, azithromycin, quinupristin, pristinamycin, virginiamycin were also found. The UOL-KIMZ-24-2 genome had genetic basis which made this

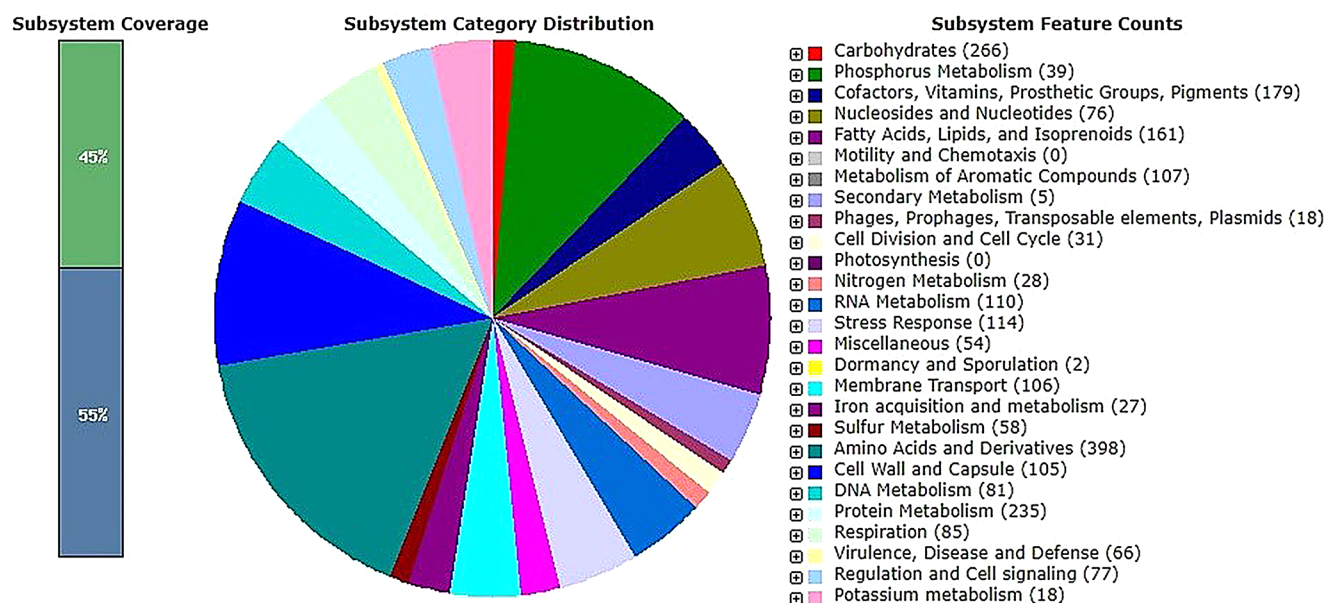


Fig. 4 Genome statistics of *A. baumannii* Strain UOL-KIMZ-24-2 through DFAST server

Table 4 Categorization of *A. Baumannii* into sequence typing group analyzed through PubMLST

Locus	Allele	Length	Location (Contig)	Start position	End position
<i>cpn60</i>	2	405	NODE_19_length_75881_cov_16.164005	48,111	48,515
<i>fusA</i>	2	633	NODE_4_length_206519_cov_15.350479	204,289	204,921
<i>gltA</i>	2	483	NODE_1_length_261381_cov_15.247610	20,822	21,304
<i>pyrG</i>	2	297	NODE_13_length_121322_cov_15.540732	28,786	29,082
<i>recA</i>	2	372	NODE_13_length_121322_cov_15.540732	102,605	102,976
<i>rpIB</i>	2	330	NODE_15_length_110889_cov_14.918293	51,333	51,662
<i>rpoB</i>	2	456	NODE_3_length_210046_cov_15.126897	203,761	204,216

strain resistant towards doxycycline, tetracycline and minocycline by *tet(B)* gene.

Mobile genetic elements (MGEs)

MGEs associated with AMR genes in *A. baumannii* strain had been predicted by VRprofile2. The output depicted that three genes *mph(E)*, *msr(E)* and *armA* encoding resistance against macrolide, macrolide, streptogramin B and aminoglycoside carried MGE belonged to IScluster/Tn. These MGE were transposases and classified into four types, scattered at different sites with AMR genes such as IS6 family transposase e.g. IS15d1, with 704 bp length and located between 121,981 and 122,685. The second MGEs was ISAb24, belonged to IS66 transposase family and lead between 122,813 and 123,946 with 1133 bp length, respectively. The third MGE was ISEc29, a member of IS4 transposase family, was present downstream to the *mph* and *msr* genes stretches, from 127,404 to 128,579 (1175 bp). However, the ISEc35 MGE (IS4 family transposase) was located next to the *arm* gene sequence that spans from 130,403 to 130,888 with 485 bp length.

Virulence-associated genes

Various virulence factors were found in the genome of *A. baumannii* UOL-KIMZ-24-2 after analyzing sequences data using VFDB database (Table S3). A total of 14 virulence factors encoded by 58 different genes (out of which 10 was found undetermined) were detected in UOL-KIMZ-24-2. The strain UOL-KIMZ-24-2 possessed genes encoding *csuABCDE* (Csu pili), *pgaABCD* ((Polysaccharide poly-N-acetylglucosamine), *adeFGH* (*AdeFGH* efflux pump/transport autoinducer), *lpsABCDLM* (lipopolysaccharide), *ompA* (Outer membrane protein), *bap* (biofilm-associated protein) and *AbaIR* (quorum sensing).

Prophage regions and genomic islands

The phage sequences were identified in genome of *A. baumannii*, predicted by PHASTEST server (Fig. 7). The genome size was 3,935,051 bp with the total 3745 genetic features, out of which 128 were phage genes and 3,617 were bacterial genes. The phage genes were localized in 3 regions in the genome of UOL-KIMZ-24-2.

Fig. 5 **A** 16 S rRNA gene based phylogeny of UOL-KIMZ-24-2 predicted by TYGS server. **B** Whole genome based phylogeny of UOL-KIMZ-24-2 predicted by TYGS server

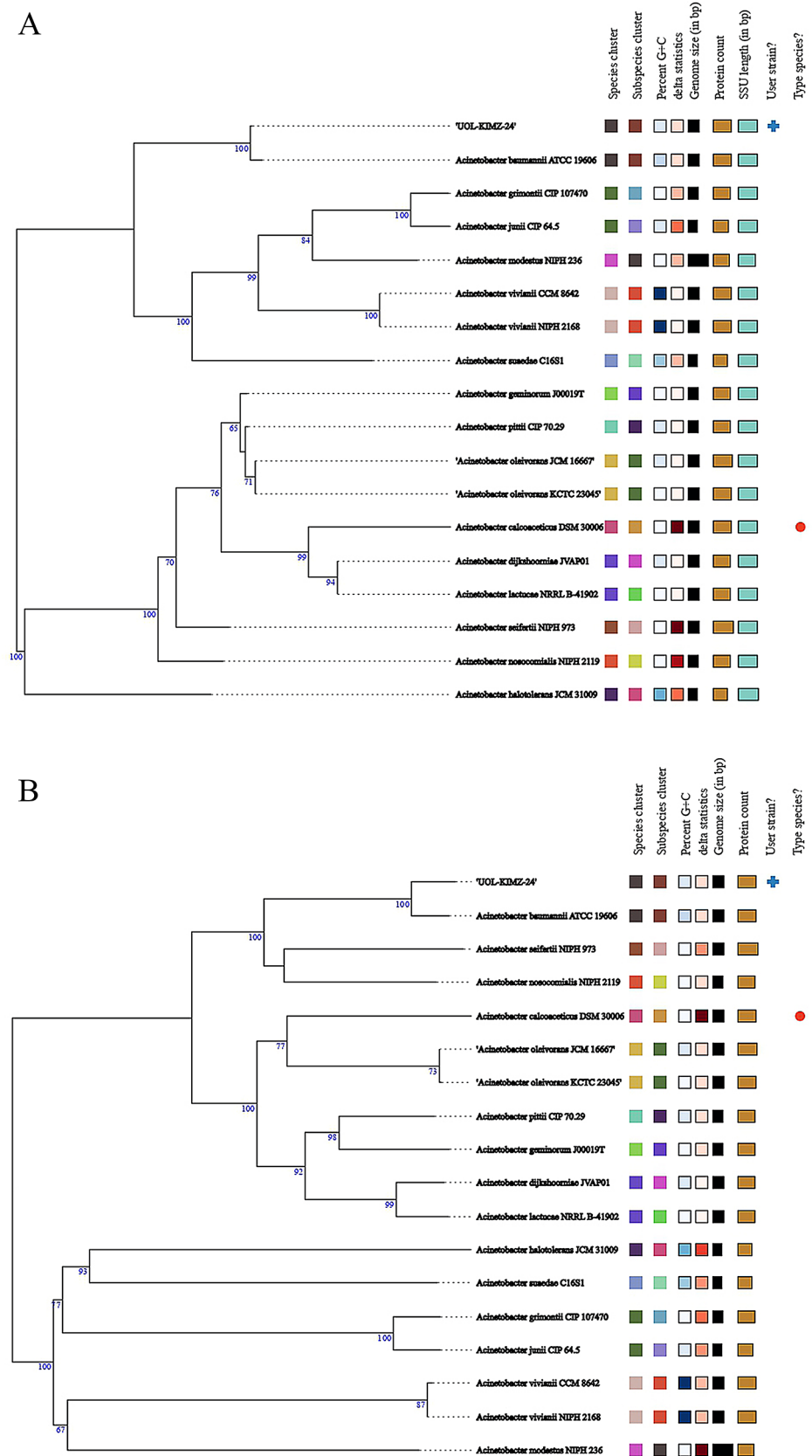


Table 5 ANI score depicting strain UOL-KIMZ-24 has the maximum similarity score with the *Acinetobacter baumannii* strain ATCC-19,606

Matric	Value
OrthoANIu %	99.47
Genome UOL-KIMZ-24 length (bp)	3,882,120
Genome <i>A. baumannii</i> length (bp)	4,011,660
Average aligned length (bp)	2,831,723
Genome UOL-KIMZ-24 coverage %	72.94
Genome <i>A. baumannii</i> coverage %	70.59

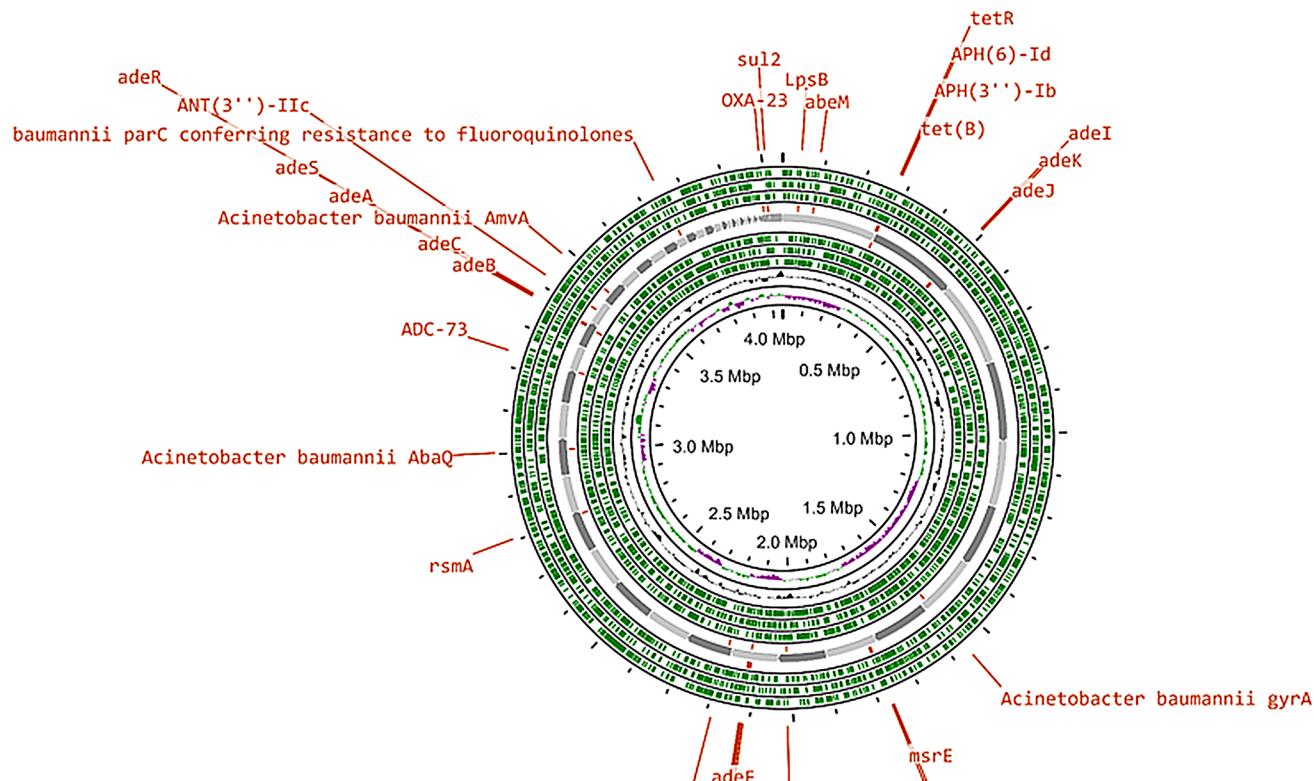
Region 1 spans between 1,488,890 and 1,679,880 with the total length of 51,682 bps, 39.36% GC content and predicted as PHAGE_Acinet_YMC11/11/R3177_NC_041866. In this region, the maximum number of genes (23) were encoding hypothetical proteins followed by phage like proteins and tail proteins (6), regulatory proteins (4), head proteins (3), transpose (1) and only one gene was involved in encoding DNA helicase, endonuclease, fiber protein and integrase (Fig. 8A).

The second bacteriophage region lies in 2,858,435 and 2,966,883 (50,243 bps), 39.2% GC content in intact form with 13 hypothetical proteins, followed by 7 phages like proteins, 5 tail proteins, 3 head proteins, 2 integrases and DNA helicase, fiber protein, holing, repressor, portal proteins and replication proteins were encoded by single gene and identified as PHAGE_Acinet_YMC11/11/R3177_NC_041866 (Fig. 8B).

However, the third region was located adjacent to region 2 flanks from 2,858,435 to 2,966,883 with the size of 28,158 bps and GC content (41.7%) and identified as PHAGE_Acinet_Bphi_B1251_NC_019541, which contained 11 genes for encoding hypothetical proteins, 7 for tail proteins, 4 for head proteins and 1 gene for phage like proteins, transpose, fiber proteins, portal proteins and 2 genes were responsible to encode terminase (Fig. 8C).

Discussion

Antibiotic resistance presents an emerging major challenge to people health, particularly advent of multi-drug resistant (MDR) bacteria poses a serious threat to health settings around the globe. Resistant to carbapenem, which is considered as last-remedy for treating multidrug-resistant Gram-negative bacteria e.g. *A. baumannii*, is spreading by previously unseen speed (O'Donnell et al. 2021). CRAB already reported resistant to several other effective antibiotics other than carbapenem (Ibrahim et al. 2021; Nang et al. 2021). This makes CRAB infections very difficult to manage by using the current available appropriate antibiotics. Hence facing these challenges, researcher and scientist around the globe are trying hard to fully understand the pattern of carbapenem resistance in *A. baumannii* strains.

**Fig. 6** Graphical circular genome map of UOL-KIMZ-24-2. The annotated antibiotic resistance genes are presented inside rings 1 and 2. The inner most ring represents GC skew pattern negative values (purple) and positive values (black)

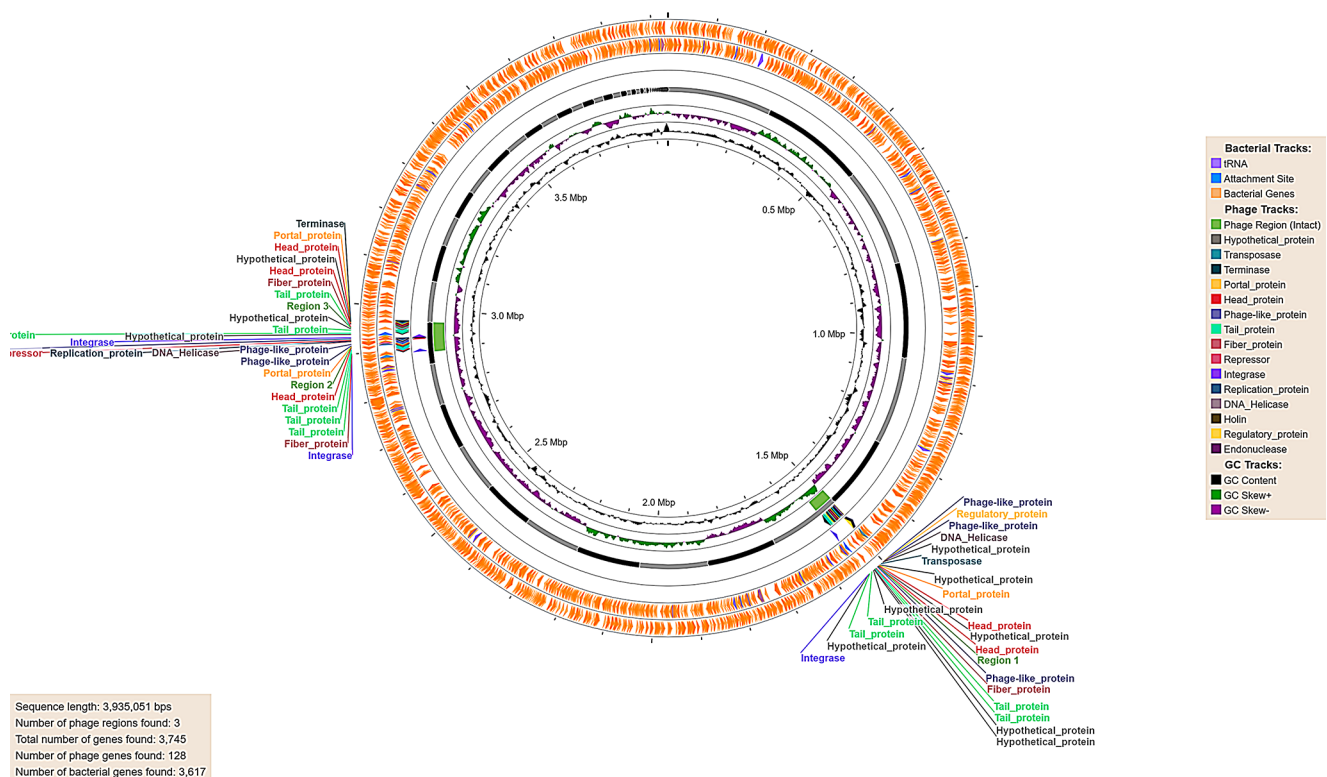


Fig. 7 A detailed overview of genome of *A. baumannii* with integrated prophages sequences / genes

Therefore, the current study was aimed to identify, evaluate whole-genome and characterize the prophages in genome of CRAB strain, recovered from patients of Lahore General Hospital, Lahore.

In the current study, a total of 150 bacterial strains were recovered from 200 samples collected from patients of Lahore General Hospital, Lahore. The highest isolates (20.6%) were recovered from endotracheal tube followed by tracheal secretions (14.6%) and wound swabs (14%). Similar observations were recorded by Begum et al. (2013) when they studied prevalence of *A. baumannii* in a Tertiary Care Hospital. In addition, the isolated strains were confirmed initially as *A. baumannii* by morphological and biochemical observations. The recovered strains were Gram-negative, coccobacilli with rounded ends as well as catalase and citrate positive and negative for oxidase and nitrate reduction. Previous studies also reported *A. baumannii* as catalase and citrate positive, negative for oxidase and nitrate reduction as well as coccobacilli with rounded ends (Mohamed et al. 2022). Along with conventional biochemical tests, the isolated strains were also confirmed as *Acinetobacter* species by API 20E system. This system is usually used to avoid the variability in findings of manual biochemical tests as identification of *Acinetobacter* species by conventional biochemical tests is difficult in the clinical microbiology laboratory (Bergogne-Berezin 2001; Al-Sehlawi et al. 2014).

In addition, assessment of antimicrobial susceptibility against 14 antibiotics was performed for all 150 *A. baumannii* strains using Kirby-Bauer method. The resistance percentages were found vary across the antibiotics classes e.g. resistance to carbapenems e.g. meropenem was found in 49.3% strains. Raut et al. (2020) reported in their study that 20% of *Acinetobacter* were resistant to meropenem, however, resistant to imipenem was in 34% strains. Our findings are also concordant with the findings reported in many other countries of Asia (Mishra et al. 2013; Sinha and Srinivasa 2007). Hence, carbapenems resistant in *Acinetobacter* species associated with nosocomial infections, mainly in an intensive care unit (ICU), reported as endemic in a tertiary care hospitals situated in the Asia-Pacific region (Banerjee et al. 2018; O'Donnell et al. 2021). In addition, *bla*_{OXA-51} genes were also detected to confirm carbapenems resistant in all 150 *Acinetobacter* species. The *bla*_{OXA-51} genes encode carbapenemases which play key role in developing carbapenems resistance in CRAB. Detecting *bla*_{OXA-51} and other related genes are widely used by several scientists for direct detection of CRAB in clinical specimens (Rafei et al. 2015; Turton et al. 2006b; Yang and Rui 2016).

In the present study, the whole-genome evaluation of *Acinetobacter baumannii* UOL-KIMZ-24 was carried out using various Bioinformatics tools for exploring presence of sequence types (STs), antimicrobial resistance (AMR)

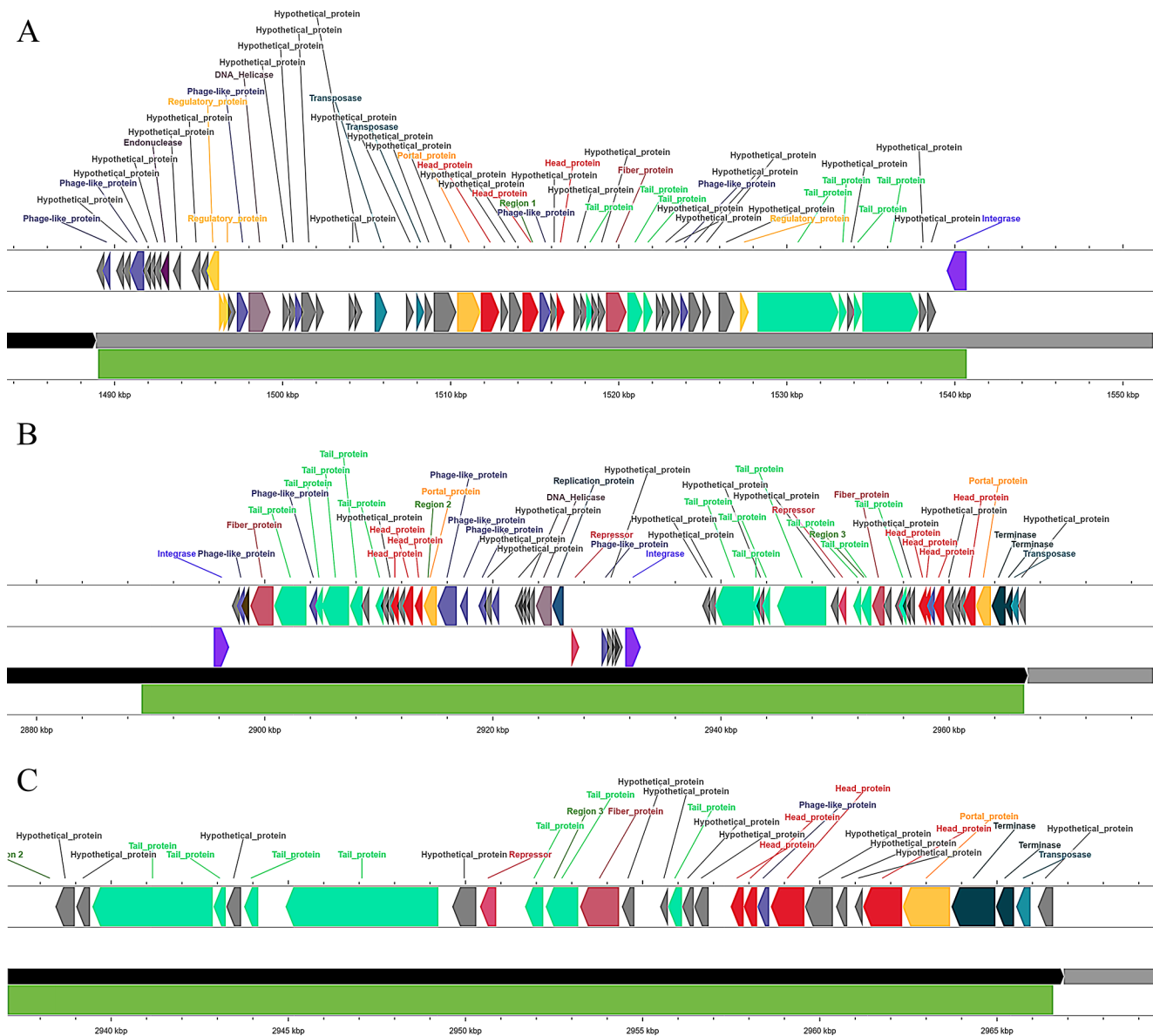


Fig. 8 A A close overview of phage region 1 in genome of *A. baumannii* UOL-KIMZ-24-2. B A close overview of phage region 2 in genome of *A. baumannii* UOL-KIMZ-24-2. C A close overview of phage region 2 in genome of *A. baumannii* UOL-KIMZ-24-2

genes, virulence genes and prophage associated with genome of UOL-KIMZ-24-2 strain. The genes such as *bla*_{OXA-66}, *bla*_{OXA-23} and *bla*_{OXA-25} were detected in the genome which are known for high resistance against imipenem, meropenem and other antibiotics belonging to β -lactum class. Among the *bla*_{OXA} genes, the *bla*_{OXA-23} is the most frequently reported genes from the genome of CRAB in many countries around the globe e.g. Egypt (Abouelfetouh et al. 2019), Germany (Eigenbrod et al. 2019), Malaysia (Rao et al. 2020), Pakistan (Khurshid et al. 2020) and Thailand (Thirapanmethee et al. 2020). The other *bla*_{OXA} variants e.g. *bla*_{OXA-66} and *bla*_{OXA-25}, were also reported in the study, which are considered as members of *bla*_{OXA-51-like} genes.

These genes possess intrinsic oxacillinase activity but lower carbapenemase activity and reported as situated naturally on the chromosome of *A. baumannii* (Abouelfetouh et al. 2019; H  ritier et al. 2005; Turton et al. 2006a). Additional to *bla*_{OXA} variants, other genes, responsible for resistance against aminoglycoside, chloramphenicol, macrolide, sulfonamide and tetracycline, were also found in the genome UOL-KIMZ-24-2 strain, hence classified as MDR strain. Previous studies also reported CRAB resistant to aminoglycosides (e.g., amikacin and gentamicin) and tetracyclines (e.g., doxycycline and tigecycline) (Begum et al. 2013; Loraine et al. 2020; Wareth et al. 2021).

Furthermore, three genes *mph(E)*, *msr(E)* and *armA* were found carrying MGEs belonged to IScluster/Tn. These MGEs were transposases and classified into four types such as *IS15d1*, *ISAb24*, *ISEc29* and *ISEc35* MGE. Chukamnerd et al. (2022) reported *ISEc29* and *ISAb24* as most frequently occurring MGEs in the genome of *A. baumannii* strains. These MGEs promote expression of many AMR genes by acting as promoters. In addition, they have also been reported for providing a cut-and-paste mechanism of transposition (Yoon et al. 2017). Based on the results related to MGEs in the current study, there are strong chances of horizontal transferring of AMR genes locate on the MGEs to other pathogenic and non-pathogenic Gram-negative bacteria in a given environment. Hence, MGEs play significant rule in acquiring and spreading of AMR genes among bacteria (Partridge et al. 2018).

In the current study, a total of 14 virulence factors encoded by 58 different genes were found in the genome of *A. baumannii* UOL-KIMZ-24-2 after analyzing sequences data using VFDB database. These virulence factors help *A. baumannii* for developing effective host-defense mechanisms (Chukamnerd et al. 2022; De Nies et al. 2021). For example, attachment and internalization into host epithelial cells of pathogens are assisted by outer membrane proteins (Harding et al. 2017), capsular exopolysaccharides help in protecting the pathogen from environmental and host-mediated stresses (Talyansky et al. 2021) and iron uptake system helps in iron chelation from the host (Greene et al. 2016). The virulence genes were associated with biofilm formation, immune evasion, enzyme, iron uptake, adherence, serum resistance and regulation. The virulence genes might be responsible for increasing the pathogenicity of UOL-KIMZ-24-2 strains as well as the sternness of infection (De Nies et al. 2021; Liu et al. 2018). Alarmingly, these genes could be transferred to other bacteria in the surrounding via horizontal gene transfer as similar as AMR genes (Chukamnerd et al. 2022; De Nies et al. 2021; Liu et al. 2018).

Moreover, the phage sequences were identified in genome of *A. baumannii*, predicted by PHASTEST server. There were three phage regions consisting of 128 genes. Region 1 and 2 were predicted as PHAGE_Acinet_YMC11/11/R3177_NC_041866. These regions were earlier already reported from the genome of carbapenem resistance of *Acinetobacter baumannii*. (Vijayakumar et al. 2022) While, the region 3 was identified as PHAGE_Acinet_Bphi_B1251_NC_019541. Similar prophage region was observed by Kenyon et al. (2019) in *Acinetobacter baumannii*. Both of the identified phages were found belonging to genus *B1251virus* and family Siphoviridae (Turner et al. 2017). The presence of these phages sequences in genome of studied strain indicates that the UOL-KIMZ-24-2 was might be previously infected with these particular phages. In addition,

these prophages might also be involved in transferring AMR genes to other bacteria by generalized transduction (Stanczak-Mrozek et al. 2017). Hence, bacteriophages are perhaps also acting as an active factor for spreading AMR or other genes.

Conclusion

In conclusion, *A. baumannii* infections were more prevalent in medical ICU followed by surgical ICU and OPD as most of the strains were isolated from these wards. Assessment of antimicrobial susceptibility revealed that resistance to carbapenems e.g., meropenem, were found in 49.3% strains. The *bla_{OXA}* genes were detected in the genome of isolated *A. baumannii* strains confirming the carbapenem resistant in *A. baumannii* strains. Whole genome analysis of *A. baumannii* UOL-KIMZ-24-2 revealed presence of various AMR genes, MGEs and virulence-associated genes as well as prophage regions, which might be transferred to other bacteria. Finally, exploring the genomic characteristics of the CRAB is important for controlling the dissemination of these pathogens in healthcare systems.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00294-024-01307-4>.

Acknowledgements The authors are very grateful to the patients for participating in this study.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection, execution, acquisition of data and analysis were performed by K. F. S. Z. H. N. F. A., S. S. A. N. S. J. M. M. A and H. A. The drafting, revising or critically reviewing the article were performed by K. F. A. S and N. H and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript. All authors gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work.

Funding No funds are available.

Data availability No datasets were generated or analysed during the current study.

Declarations

Consent for publication All authors have read and approved the final.

Competing interests The authors declare no competing interests.

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